Renke Wang

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Professional Summary

Ph.D. candidate in Electrical and Computer Engineering specializing in control systems, mathematical optimization, resource-constrained scheduling, resource allocation, routing and path planning, and real-time decision-making. Develops optimization, control, search, and learning-based methods for networked and multi-agent systems, with experience in Model Predictive Control (MPC), robust control and stability analysis, distributed optimization, scheduling and control co-design, event-driven simulation, and computational benchmarking.

Education

George Mason University
Ph.D. Candidate in Electrical and Computer Engineering — *Expected 2027*

Dalian University of Technology
Master of Engineering in Control Science and Engineering

Dalian University of Technology
Bachelor of Engineering in Automation

Core Skills

- **Mathematical Optimization and Operations Research:** Linear Programming (LP), Mixed-Integer Programming (MIP), Mixed-Integer Linear Programming (MILP), Nonlinear Programming (NLP), Mixed-Integer Nonlinear Programming (MINLP), and nonconvex constrained optimization; centralized and distributed optimization; Alternating Direction Method of Multipliers (ADMM) and Consensus Alternating Direction Method of Multipliers (C-ADMM).
- **Scheduling and Planning:** Resource-constrained, real-time, and priority-based scheduling; resource allocation; task allocation; task and resource-occupation sequencing; priority assignment and dispatching; routing and shortest-path planning; joint path planning and scheduling; online decision-making and multi-agent coordination under shared-resource constraints.
- **Algorithms, Modeling, and Analysis:** Decision-tree search, graph-search methods, Dijkstra's algorithm, A* search, multi-step lookahead, event-driven simulation, Monte Carlo simulation, scenario and sensitivity analysis, model validation, baseline benchmarking, solver diagnostics, computational performance evaluation, and runtime and scalability analysis.
- **Control Systems and Stability Analysis:** State-space modeling, feedback controller design, optimal control, Model Predictive Control (MPC), robust MPC, tube-based MPC, scheduling and control co-design, Linear Matrix Inequality (LMI)-based controller synthesis, Lyapunov-based stability analysis, asymptotic stability, exponential stability, stochastic stability, positive systems, switched systems, Markov Jump Linear Systems (MJLS), dwell-time methods, event-triggered control, and robust control under timing uncertainty.
- **Machine Learning and Sequential Decision-Making:** Supervised neural policy learning, Deep Reinforcement Learning (DRL), actor-critic methods, Proximal Policy Optimization (PPO), learning from exact optimization and search decisions, and neural policy evaluation.

Tools and Technologies

- **Programming and Data Analysis:** Python, MATLAB, Pandas, NumPy, SciPy, Matplotlib, and Microsoft Excel.
- **Optimization and Machine Learning:** Gurobi, CasADi, YALMIP, MATLAB Optimization Toolbox, and PyTorch.
- **Development and Computing:** Git, GitHub, Linux, Jupyter Notebook, and LaTeX.

Research Experience

George Mason University — **Fairfax, VA**
Research Assistant — *August 2022 – Present*

Multi-Agent Spatial-Resource Scheduling and Coordination

- **Optimization and Scheduling:** Developed a Contention-Resolving Model Predictive Control (CR-MPC) framework for shared-resource access sequencing and occupation timing in multi-agent systems; applied the framework to warehouse merging zones, multi-lane intersections, and multi-intersection traffic networks, and benchmarked scheduling performance against priority-based methods.

- **Simulation and Visualization:** Built event-driven browser-based simulations and animations to evaluate agent trajectories, resource utilization, contention-resolution decisions, scheduling performance, and computational behavior across different operating scenarios.

Distributed Scheduling and Control Co-Design

- **Distributed Optimization:** Developed a Consensus ADMM-based framework that decomposes multi-layer scheduling, sequencing, and resource-allocation problems into resource-specific subproblems while coordinating task timing through consensus constraints.
- **Benchmarking and Scalability:** Implemented parallel local optimization workflows and evaluated solution performance, computation time, and scalability against centralized optimization and established priority-based scheduling methods.

Scheduling and Control Co-Design under Heterogeneous Resources

- **Heterogeneous Resource Scheduling:** Developed a centralized contention resolving MPC framework for joint scheduling, resource allocation, and control under heterogeneous resource constraints; evaluated the framework through Monte Carlo simulations for multiple lighter-than-air agents and extended the scheduling methodology to multi-robot coordination and obstacle-avoidance applications.
- **Human–Multi-Robot Decision Support:** Developed the core real-time scheduling methodology for an attention-guided recommendation system that determines when recommendations should be issued and which actions should be recommended to a human operator supervising multiple robots; collaborated with the project team on system integration and experimental evaluation.

Learning-Based Path Selection and Shared-Resource Scheduling

- **Supervised Policy Learning:** Developed a neural scheduling policy trained on optimal decisions generated by exact contention-resolving MPC decision-tree search, using optimization-generated decisions to learn event-driven contention-resolution policies under shared-resource constraints.
- **Deep Reinforcement Learning:** Developing an event-driven Proximal Policy Optimization (PPO) actor-critic framework for joint path selection and shared-resource scheduling, extending fixed-route scheduling to online routing and priority decisions for multi-agent systems.

Robust Control under Distribution-Free Timing Perturbations

- **Linear Systems and Stability:** Developed a tube-based Model Predictive Control (MPC) framework for Linear Time-Invariant (LTI) systems under distribution-free control-update timing perturbations; designed timing-dependent dynamic feedback gains, derived the resulting error dynamics, and established asymptotic stability and the existence of a forward invariant set.
- **Nonlinear Systems and Data-Driven Koopman Modeling:** Koopman operator-based lifted linear modeling of nonlinear system dynamics and data-driven identification of Koopman operators from trajectory data, with applications to Model Predictive Control (MPC) and robustness analysis under timing perturbations.

Dalian University of Technology — **Dalian, China**
Graduate Researcher — *September 2017 – June 2020*

- **System Modeling and Stability Analysis:** Derived Lyapunov-based stability conditions for positive switched systems and Markov Jump Linear Systems (MJLS) under dwell-time constraints, stochastic mode transitions, and fully or partially known transition rates.
- **Controller Design:** Designed state-feedback and L1 controllers using piecewise-linear copositive Lyapunov functions, establishing stochastic and exponential stability and L1-gain performance under disturbances and controller failures.

Selected Publications and Manuscripts

- Renke Wang and Ningshi Yao. "Distributed Scheduling and Control Co-Design for Networked Control Systems with Multi-Layered Shared Resources." *IEEE Transactions on Control of Network Systems (TCNS)*, under review.
- Renke Wang, Mengxue Hou, and Ningshi Yao. "Robust Co-Design of Scheduling and Control for Lighter-than-Air Agents with Heterogeneous Resources." *International Journal of Robust and Nonlinear Control*, under review.
- J. Lian and R. Wang. "Stochastic Stability of Positive Markov Jump Linear Systems with Fixed Dwell Time."

Nonlinear Analysis: Hybrid Systems, 40 (2021): 101014.

– R. Wang and N. Yao. "Contention-Resolving Model Predictive Control for Coordinating Automated Vehicles at a Multi-Lane Traffic Intersection." *2025 IEEE Conference on Control Technology and Applications (CCTA)*. IEEE, 2025.

– Rajul Kumar, Renke Wang, and Ningshi Yao. "One Human and Multi-Robot Collaboration: Evaluating the Impact of Attention-Guided On-Screen Recommendations on Human Multitasking Efficiency, Cognitive Load, and Trust." *2025 34th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*. IEEE, 2025.

– R. Wang and N. Yao. "Robust Model Predictive Control for Networked Control Systems with Timing Perturbations." *2024 American Control Conference (ACC)*. IEEE, 2024.

– N. Yao and R. Wang. "Optimal Real-Time Human Attention Allocation and Scheduling in a Multi-human and Multi-robot Collaborative System." *2024 IEEE Conference on Control Technology and Applications (CCTA)*. IEEE, 2024.

Teaching Experience

George Mason University **Fairfax, VA**

Teaching Assistant *August 2022 – December 2022*

– Delivered lectures and led or co-led laboratory sessions for Continuous-Time Signals and Systems; explained technical concepts, guided laboratory work, graded assignments, and supported undergraduate students.

– Graded coursework and answered student questions for Classical Systems and Control Theory, providing clarification on course concepts and assignments.

Honors and Professional Service

– Summer Research Assistantship Award, George Mason University Graduate Division, 2026.

– Reviewer for *IEEE Transactions on Control Systems Technology*; *IEEE Transactions on Systems, Man, and Cybernetics: Systems*; and IEEE conferences including ACC, CDC, and CCTA, 2023–Present.